3.4 Algebraic Topology — Problems

Work these alongside the schedule. Full solutions are in solutions.md; try each problem for at least 30 minutes before looking. You may quote standard results from Hatcher (Chapters 0–2, §§3.1–3.2) unless the problem asks you to prove them.

Warm-up

1. Show that S^n is simply connected for $n \geq 2$. (Use van Kampen with two open sets.)

2. Prove there is no retraction $r : D^2 \to S^1$, and deduce that every map $h : D^2 \to D^2$ has a fixed point.

3. (a) Compute $\pi_1(T^2)$ and $\pi_1(S^1 \vee S^1)$.

(b) Show that T^2 and $S^1 \vee S^1$ are not homotopy equivalent, using π_1 only.

4. Use the Mayer–Vietoris sequence to compute $\tilde{H}_k(S^n)$ for all k and $n \geq 1$, by induction on n.

5. Using the standard CW structure, compute $\chi(M_g)$ for the closed orientable surface of genus g and $\chi(N_h)$ for the connected sum of h projective planes. For which g, h do the Euler characteristics agree? Show that even in those cases M_g and N_h are not homotopy equivalent, using H_1 (computed from the same CW structures).

6. Let X be contractible. Show that $\tilde{H}_n(X) = 0$ for all n and that $\pi_1(X, x_0) = 0$ for all x_0.

Core

7. (a) Show that the quotient map $S^2 \to \mathbb{RP}^2$ is a 2-sheeted covering, and deduce $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2$.

(b) Show that every map $\mathbb{RP}^2 \to S^1$ is nullhomotopic.

8. The Klein bottle K is the square $[0,1]^2$ with the identifications $(0, y) \sim (1, y)$ and $(x, 0) \sim (1 - x, 1)$.

(a) Give a CW structure on K and use it to show $\pi_1(K) \cong \langle a, b \mid abab^{-1} \rangle$.
(b) Show that $\pi_1(K)$ is non-abelian.
(c) Compute $H_1(K)$ from (a).

9. Compute $H_k(\mathbb{RP}^n; \mathbb{Z})$ for all k using cellular homology. State clearly the degree of the relevant attaching maps and justify it.

10. (a) Show that the antipodal map $a : S^n \to S^n$ has degree $(-1)^{n+1}$.

(b) Prove the hairy ball theorem: S^n admits a nowhere-vanishing continuous tangent vector field if and only if n is odd.

11. Prove Brouwer's fixed point theorem for D^n, $n \geq 1$, using homology.

12. Describe all connected 2-sheeted covering spaces of $X = S^1 \vee S^1$ up to isomorphism. For each, identify $p_*\pi_1$ as a subgroup of $F_2 = \langle a, b \rangle$ and say whether the cover is normal.

13. Let $m \geq 2$ and q be coprime to m. Let $\mathbb{Z}/m$ act on $S^3 \subset \mathbb{C}^2$ by $k \cdot (z, w) = (\zeta^k z, \zeta^{kq} w)$ with $\zeta = e^{2\pi i/m}$. Show that this is a covering space action and that $L(m, q) = S^3/(\mathbb{Z}/m)$ has $\pi_1 \cong \mathbb{Z}/m$.

14. Use Mayer–Vietoris to compute $H_*(K)$ for the Klein bottle K, viewing K as the union of two Möbius bands glued along their boundaries.

15. Show that if $\mathbb{R}^m$ and $\mathbb{R}^n$ are homeomorphic then $m = n$. (Use local homology.)

Qualifying-exam level

16. Let $A \subseteq X$ be a retract of X, with retraction r and inclusion i.

(a) Show $i_* : H_n(A) \to H_n(X)$ is injective and $H_n(X) \cong H_n(A) \oplus H_n(X, A)$.
(b) Deduce that the Möbius band M does not retract onto its boundary circle.

17. (a) Show that every map $f : S^2 \to T^2$ is nullhomotopic.

(b) Construct a map $g : T^2 \to S^2$ that induces an isomorphism on H_2.
(c) Deduce that for any maps $f : S^2 \to T^2$ and $g : T^2 \to S^2$, the composite $g \circ f : S^2 \to S^2$ has degree 0.

18. Using the cup product structure of $H^*(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^3)$, $|x| = 2$, show that $\mathbb{CP}^2$ and $S^2 \vee S^4$ have isomorphic homology groups but are not homotopy equivalent.

19. (a) Show that a connected covering space of a connected graph is a graph, and deduce that every subgroup of a free group is free.

(b) Let $H \leq F_n$ have finite index k. Show that H is free of rank $k(n-1)+1$.
(c) Deduce that F_2 contains a subgroup isomorphic to F_3.

20. Let $f : S^n \to S^n$ be a map. Show:

(a) if f has no fixed point, then $\deg f = (-1)^{n+1}$;
(b) if $f(x) \neq -x$ for all x, then $\deg f = 1$;
(c) if n is even, then for every map $f : S^n \to S^n$ there is x with $f(x) = x$ or $f(x) = -x$. Deduce that every map $\mathbb{RP}^{2k} \to \mathbb{RP}^{2k}$ has a fixed point (you may use that a map $\mathbb{RP}^n \to \mathbb{RP}^n$, $n \geq 2$, lifts to a map $S^n \to S^n$).

21. Show that there is no map $g : S^2 \to S^1$ with $g(-x) = -g(x)$ for all x. Deduce the Borsuk–Ulam theorem for $n = 2$: every map $f : S^2 \to \mathbb{R}^2$ has some x with $f(x) = f(-x)$.